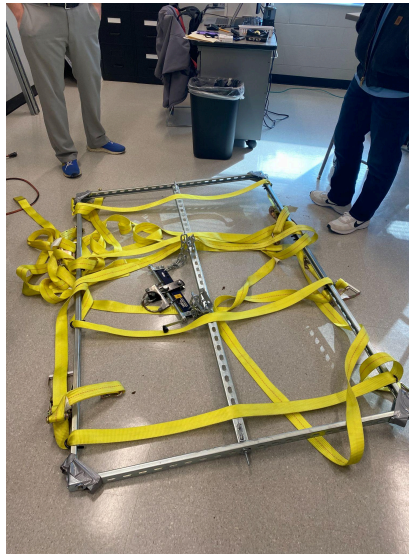


Element K-

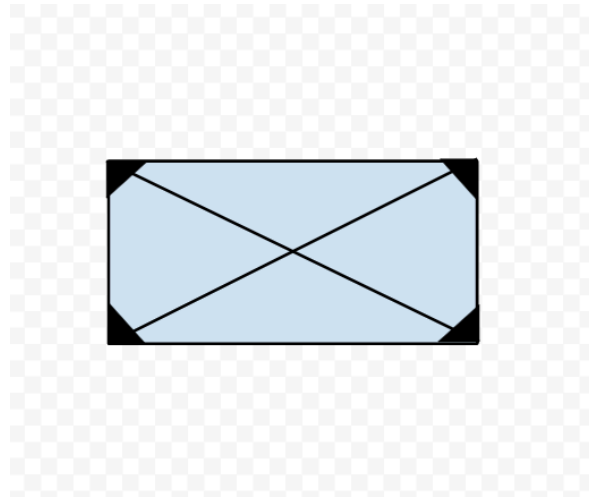
Reason for Coffin-Style Bracing

We ultimately went with coffin-style bracing. The first year they used a single I-beam bracing technique with a thin layer of duct tape over already weak corner bracing attached. This resulted in the frame completely buckling on test day over merely the weight of one person due to the inadequate bracing. This project laid the foundation for the years to come making this attempt far from a failure. That group also provided the materials for the second year attempt as well as ours.

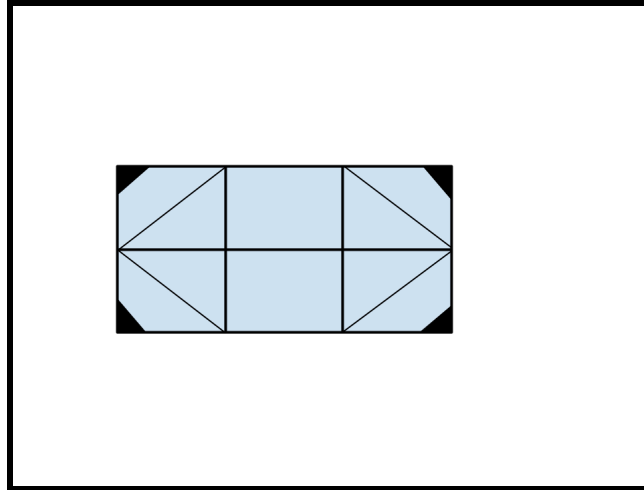


The second year's bracing was certainly an improvement upon the year prior. They implemented an X-bracing which better distributed the weight among the frame. Although the distribution was an improvement, the integrity of the frame as a whole improved little if not nothing at all. There were other ways this group could have approached the stability problem, and integrity was the biggest flaw that the bracing pertained to in the first place. While this bracing technique wasn't a massive success, it

gave us valuable knowledge and insight as to what type of bracing we would ultimately end up with.



Finally, for our bracing, we went with our own coffin-style bracing. This decision came from multiple attempts at small prototypes as well as consideration from the previous two attempts. We constructed multiple different popsicle stick models of potential frame options, including the first and second bracing techniques as shown above. We put these small models through simple stress tests to help show us which ones were the best for our project. We also took in consideration from the previous attempts and realized that we would need far more support such as beams to improve the integrity of the frame. The coffin-style frame was the most successful of all of our miniature prototypes by far, so we took it and ran with it. We are confident that it can handle the intended weight of 2000 lbs because of the stress test we put the frame through carrying a 1-ton weight. It bent less than one inch total and had absolutely no remaining distortion once the weight was removed.



Reason for Electric Motor

In our project, the metal frame helps support the weight of the animals while also carrying them across an operating room. The frame is suspended from a 2-ton capable motor that moves the frame along the track. The main reason for the addition of our own electric motor is for stability more than anything. The purpose of the electric motor is to be able to stabilize both of our engine levelers in real time. With having two engine levelers, being able to adjust them in real time conveniently is very nice, but also very important. If an animal were to make a sudden movement that could be quickly countermeasured by simply clicking a button that will adjust both engine levelers simultaneously.

Reason for Second Engine Leveler and Shepard's Crook

For years prior, there has been only one engine leveler for this lift. While an engine leveler is very helpful, it is rendered useless if not easily adjustable at a moment's notice. Now that we implemented our own electric motor, engine levelers can be used for their intended purpose and serve at full functionality. After watching videos

of how the engine leveler addressed the tilt on both axes, we determined it would be better to have two engine levelers. By having two engine levelers, we automatically address any tilt in one direction, and also are able to address the other direction by the use of our electric motor to adjust the frame in real time. With only one engine leveler positioned over the center of the frame connected from the motor, it only addressed one axis and therefore had faulty balancing. Since we added another engine leveler, we made another trolley that rests on the track that follows the main engine that has one of the engine levelers suspended from it. This allows for the first engine leveler to hang from the original motor and be attached to the front of the frame, while the second engine leveler is attached to the trolley and the back of the frame. To make sure that the trolley stays at the same position in relativity to the main motor, we have a precisely measured shepard's crook that serves as a hook to keep it secure at a set distance apart from the motor.

Reason for Welded Joints Instead of Bolts

The previous two groups ended up going with the use of bolts to secure the frame. They used bolts to secure the I-beam in place the first year, and they used bolts again to secure the X-beam in place last year. Both years there was clear distortion of the frame regardless of the I-Beam or X-Beam. They were both inadequate framing techniques, and one major reason for that is the use of bolts instead of welding. Bolts arent as secure as a decent welding job, and our team welder Corbin Wolfgang did an excellent job. With his welding experience, welding became a feasible alternative to the ever so faulty bolts. With the use of welding, this also gave our group options for way

more creative frame concepts, including our coffin-style bracing that we ended up going with.